

Problem C. Deletions of Two Adjacent Letters

Time Limit 2000 ms

Mem Limit 262144 kB

The string s is given, the string length is **odd** number. The string consists of lowercase letters of the Latin alphabet.

As long as the string length is greater than 1, the following operation can be performed on it: select any two adjacent letters in the string s and delete them from the string. For example, from the string "lemma" in one operation, you can get any of the four strings: "mma", "lma", "lea" or "lem". In particular, in one operation, the length of the string reduces by 2.

Formally, let the string s have the form $s = s_1 s_2 \dots s_n$ ($n > 1$). During one operation, you choose an arbitrary index i ($1 \leq i < n$) and replace $s = s_1 s_2 \dots s_{i-1} s_{i+2} \dots s_n$.

For the given string s and the letter c , determine whether it is possible to make such a sequence of operations that in the end the equality $s = c$ will be true? In other words, is there such a sequence of operations that the process will end with a string of length 1, which consists of the letter c ?

Input

The first line of input data contains an integer t ($1 \leq t \leq 10^3$) — the number of input test cases.

The descriptions of the t cases follow. Each test case is represented by two lines:

- string s , which has an odd length from 1 to 49 inclusive and consists of lowercase letters of the Latin alphabet;
- is a string containing one letter c , where c is a lowercase letter of the Latin alphabet.

Output

For each test case in a separate line output:

- YES, if the string s can be converted so that $s = c$ is true;
- NO otherwise.

You can output YES and NO in any case (for example, the strings yEs, yes, Yes and YES will be recognized as a positive response).

Examples

Input	Output
5 abcde c abcde b x y aaaaaaaaaaaaaaaa a contest t	YES NO NO YES YES

Note

In the first test case, $s = \text{"abcde"}$. You need to get $s = \text{"c"}$. For the first operation, delete the first two letters, we get $s = \text{"cde"}$. In the second operation, we delete the last two letters, so we get the expected value of $s = \text{"c"}$.

In the third test case, $s = \text{"x"}$, it is required to get $s = \text{"y"}$. Obviously, this cannot be done.